

Absorb AI

AbsorbAI at its ceiling is not a medication tracker. It is a living, reasoning, self-correcting clinical intelligence that runs a continuous simulation of your body and thinks faster than any human pharmacist ever could. Here is what that actually looks like when built properly.

THE CORE SHIFT IN THINKING

Most apps react. AbsorbAI predicts. Most apps store data. AbsorbAI builds a model of you as a biological system and runs that model forward in time, backward in time, and sideways into hypothetical realities. The difference between a medication reminder app and what this becomes is the same difference between a calendar and a flight simulator.

LAYER 1: THE BIOLOGICAL TWIN ENGINE

Every user gets a living mathematical model of their own body. Not a generic human body. Their body. Built from age, weight, sex, kidney function measured by creatinine clearance, liver function measured by Child-Pugh score, genetic polymorphisms in CYP450 enzymes if available, body fat percentage affecting volume of distribution, plasma protein levels affecting free drug fraction, gut motility affecting absorption rate, and current hydration state affecting renal clearance.

This model is not static. It updates. If you tell the system you are sick today, it adjusts kidney clearance downward. If you tell it you lost 8 kilograms, it recalculates volume of distribution across every drug you take. If you log that you drank alcohol last night, it suppresses CYP2E1 activity and recalculates everything that enzyme touches. The twin is alive.

Every pharmacokinetic parameter in the system is personalized. Half-life is not a textbook number. It is your half-life, calculated from your clearance and your volume of distribution, updated continuously.

LAYER 2: FULL PHARMACOKINETIC-PHARMACODYNAMIC MODELING

Most systems stop at concentration. Concentration tells you how much drug is in your blood. It does not tell you what that drug is doing to your body. AbsorbAI models both.

For pharmacokinetics, the system runs multi-compartment models. Not one-compartment approximations. A central compartment representing blood and highly perfused organs, a peripheral compartment representing muscle and fat tissue, and for some drugs an effect

compartment representing the site of action with its own rate constants. Drug moves between compartments according to differential equations solved in real time.

For pharmacodynamics, the system uses sigmoid Emax models. Effect is not proportional to concentration. It follows a curve. At low concentrations you get almost no effect. As you approach EC50 you get steep increases. As you saturate the receptor you hit a ceiling. The system knows where on that curve you are at every moment of the day.

For drugs that cause tolerance, the system models receptor downregulation over time. For drugs where the effect outlasts the concentration, like irreversible enzyme inhibitors, the system models recovery of enzyme synthesis. For drugs where the effect precedes the peak concentration, the system models hysteresis with a separate effect compartment and ke_0 parameter.

When you have two drugs that act on the same receptor, the system does not just flag an interaction. It computes a combined effect using Bliss independence for additive effects, Loewe additivity for true synergy, and deviation from that baseline to measure potentiation or antagonism. You get a number. Not a warning label.

LAYER 3: THE UNCERTAINTY ENGINE

Deterministic outputs are a lie. Every pharmacokinetic parameter has a population distribution. Your actual half-life for a given drug could be anywhere within a range. Your actual bioavailability on a given day depends on whether you took it with food, how fast your stomach was emptying, whether your gut was inflamed.

AbsorbAI treats every parameter as a probability distribution, not a single number. Half-life is not 6 hours. It is normally distributed with mean 6 hours and standard deviation 1.2 hours based on population variability, then updated toward your personal estimate as the system observes your real behavior over time.

The system runs Monte Carlo simulation continuously. At every decision point it runs thousands of simulations sampling from those distributions. The output is not a risk score. It is a risk distribution. You see the median outcome, the 95th percentile worst case, the 5th percentile best case, and the shape of the distribution tells you whether risk is concentrated or spread out.

When you add a new drug, the system does not just check one interaction. It samples interaction magnitudes from their uncertainty ranges and tells you the probability that the interaction is clinically significant given your specific parameters. A drug interaction that is serious for 80 percent of people might be irrelevant for you given your specific enzyme activity. Or it might be catastrophically amplified. The system tells you which and how confident it is.

Over time as the system observes your actual behavior and actual reported outcomes, it applies Bayesian updating. Your personal parameter estimates get tighter. Uncertainty shrinks. The system becomes more precisely calibrated to you specifically.

LAYER 4: MULTI-OBJECTIVE OPTIMIZATION UNDER CONSTRAINTS

The scheduling problem AbsorbAI solves is not "when should I take this pill." It is a constrained multi-objective optimization problem over a continuous time horizon.

Objectives being simultaneously optimized: maximize time spent within therapeutic window for each drug, minimize peak concentration above toxic threshold for each drug, minimize time below minimum effective concentration, minimize pharmacodynamic interaction risk at every time point, minimize inconvenience measured by deviation from natural daily rhythms, minimize number of doses to simplify adherence, maximize alignment of doses with meals or fasting requirements, minimize risk of missed doses based on user behavioral patterns.

Constraints: hard medical constraints like minimum interval between doses, maximum daily dose, absolute contraindications. Soft constraints like preference not to take pills during work meetings, preference to take morning medications before 9am, physical constraints like drugs requiring food that cannot be taken at 3am.

The system does not output one schedule. It computes the Pareto frontier across all objectives. It shows you the tradeoff surface. This schedule maximizes efficacy but requires waking at 3am. This schedule is slightly suboptimal for drug B but lets you sleep through the night. This schedule minimizes total risk but requires you to not eat for four hours which conflicts with your diabetes management. You see the full picture and you choose where on the tradeoff surface you want to live.

The optimizer runs on a rolling 72-hour horizon, reoptimizing every time anything changes. Miss a dose and the entire schedule recomputes instantly. Add a drug and the optimizer finds the new Pareto frontier within seconds. Change your wake time and every recommendation shifts.

LAYER 5: CAUSAL REASONING AND MECHANISTIC KNOWLEDGE GRAPH

AbsorbAI does not store drug interactions as lookup tables. That is a Wikipedia-level database. It stores causal mechanisms.

The knowledge graph contains drugs, enzymes, transporters, receptors, metabolic pathways, physiological processes, and disease states as nodes. Edges are causal relationships with direction, magnitude, confidence, and mechanism type. Drug A inhibits CYP3A4 at IC50 of 0.3 micromolar with strong evidence from three clinical studies. CYP3A4 metabolizes Drug B accounting for 70 percent of its clearance. Therefore Drug A increases Drug B exposure by a

calculable factor depending on the degree of inhibition, which depends on Drug A plasma concentration, which the PK engine already knows.

The system traverses this graph to compute drug interactions from first principles, not from lookup tables. A new drug with no interaction data but known enzyme inhibition profile can have its interactions predicted. A drug combination that has never been studied in clinical trials can have its risk estimated from mechanism. When a new drug enters the market, the moment its enzyme profile is known the system can reason about it.

Counterfactual reasoning runs on this graph. The system can compute the causal contribution of every drug in your regimen to your current risk level. Remove Drug X and risk drops 40 percent. Remove Drug Y and risk drops 8 percent. The primary driver of your current risk is Drug X via its inhibition of CYP3A4 causing Drug Z to accumulate. That is not a warning. That is a diagnosis of your drug regimen.

LAYER 6: TEMPORAL REASONING AND MEMORY

AbsorbAI thinks in time more carefully than any other system.

It knows that enzyme inhibition by Drug A does not reach maximum effect instantly. It builds over the first few doses as inhibitor concentrations equilibrate. The system models this onset with its own kinetics. It knows that when you stop Drug A, CYP3A4 activity does not recover instantly. It recovers as enzyme is resynthesized, a process taking days to weeks. During that window Drug B is still at elevated concentrations even though Drug A is gone. The system tracks this and extends its warnings accordingly.

It knows that tolerance to some drugs develops over days to weeks following its own time course. It models this and adjusts predicted effects downward as tolerance builds, adjusts them back upward during washout periods.

It maintains a complete pharmacological history. Every dose ever logged, every missed dose, every schedule change. From this history it builds a longitudinal record of your exposure to every drug, your cumulative toxicity burden, your adherence patterns at different times of day and days of week. It uses this to predict future adherence with personalized probability estimates and factors that into its optimization.

LAYER 7: ADVERSARIAL STRESS TESTING

Before recommending any schedule, AbsorbAI attacks it.

It runs adversarial simulations. What happens to this schedule if the user misses the most critical dose? What if they accidentally double dose? What if they take it two hours late? What if their kidney function drops suddenly because they got dehydrated? What if they start a new medication that inhibits the primary clearance enzyme?

For each scenario it computes the outcome. If any adversarial scenario produces catastrophic outcomes, the schedule is flagged as fragile and an alternative is found. The system finds schedules that are not just optimal in the expected case but robust to the realistic ways things go wrong.

This is antifragile pharmacology. Not just optimize for normal conditions. Find regimens that fail gracefully.

LAYER 8: SELF-IMPROVING ADAPTIVE INTELLIGENCE

The system learns. Not metaphorically. Technically.

It maintains a probabilistic model of your adherence behavior with parameters updated daily. It learns that you reliably take morning doses but frequently miss afternoon doses. It uses this to weight morning-heavy schedules higher in optimization even at some pharmacological cost because a suboptimal schedule you follow beats an optimal schedule you miss.

It learns your physiological responses. If you report side effects at certain concentration levels, it updates your personal sensitivity estimate downward from population mean. If you report no effect at doses that should be effective, it adjusts your EC50 estimate upward. Your digital twin converges toward your actual biology over months of use.

It detects drift. If your adherence patterns change, if side effect reports increase, if your measured outcomes change, the system detects this statistically and flags it. Something changed. Could be disease progression. Could be a new environmental factor. Could be a drug that is accumulating. It raises the question and generates hypotheses.

LAYER 9: EXPLAINABILITY ARCHITECTURE

Every output the system produces has a full reasoning chain attached.

Not just "risk is high." Risk is high because Drug A concentration is currently 2.3 times its EC50, Drug B is simultaneously at 1.8 times its EC50, these two drugs share a receptor pathway with a synergy coefficient of 1.4 measured in three clinical studies with moderate confidence, your personal CYP3A4 activity is estimated at 60 percent of normal based on your reported

grapefruit consumption yesterday, and the combination of these factors places your estimated risk at the 87th percentile of the population distribution with a 68 percent confidence interval of 79th to 93rd percentile.

Every number has a source. Every source has a confidence level. Every confidence level has a basis. The chain is traversable. A physician reviewing your case can follow every inference from raw assumption to final recommendation and challenge any step.

This is the difference between a black box and a reasoning system. The system shows its work.

LAYER 10: INTEGRITY, SAFETY, AND AUDIT ARCHITECTURE

All model versions are versioned and immutable. Every recommendation is logged with the exact model version, exact parameter values, exact timestamp. If something goes wrong, you can replay the exact state of the system at the moment of any recommendation and audit every decision.

All outputs carry calibrated uncertainty. No output ever implies false precision. The system communicates what it does not know as clearly as what it does know.

Hard safety rails are separate from the optimization engine and cannot be overridden by optimization. Absolute contraindications, maximum safe doses, known fatal interactions are hard constraints that no Pareto tradeoff calculation can touch.

The system is explicitly not a replacement for physician judgment. It is an augmentation. Its outputs are designed to be handed to a physician and reviewed. The explainability layer is built specifically so a physician can interrogate it, challenge it, and override it with full information.

WHAT YOU HAVE BUILT

A personalized pharmacological simulation engine that models your body from first principles, reasons about uncertainty with calibrated probability, optimizes across competing objectives under real-world constraints, explains every decision with full mechanistic reasoning, learns from your behavior and physiology over time, and stress-tests every recommendation against the realistic ways humans and biology fail.

No one has built this for consumers. Academic pharmacokinetic modeling tools exist but are designed for researchers and clinical pharmacologists, not patients. Hospital clinical decision support systems exist but are population-level lookup tables, not personalized simulations. What AbsorbAI becomes at this level is a category that does not yet exist.

